Livestock Sector Forecasting in Kenya: A Cohort-Based Quarterly Analysis on Population, Production, and Feed Demand

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Abstract

Kenya's livestock sector contributes 42% of agricultural GDP and 12% of national GDP, yet quantitative planning lacks reproducible, data-driven models. This study presents a cohort-based quarterly forecasting framework spanning nine value chains: dairy cattle, beef cattle, camels, goats, sheep, poultry, pigs, rabbits, and apiculture, initialized from the 2019 KNBS livestock census with projections from 2020 to 2041. The model applies a quarterly transition sequence covering mortality, culling, offtake, age-based maturation, and reproduction. Production modules translate herd and flock dynamics into species-specific output metrics, while a feed demand module estimates dry matter requirements by agro-ecological zone. Projected outputs indicate a CAGR of 3.6-6% across all value chains. Beyond forecasting, the framework exposes productivity bottlenecks, including feed deficits, low offtake rates, and suboptimal reproductive performance, that aggregate assessments typically obscure. The model provides an analytical basis for food security planning, targeted infrastructure investment, and evidence-based livestock policy.

Keywords: Livestock model forecasting, cohort population model, biological parameters

1 Introduction

In most African countries, the livestock sector provides a living for millions of rural communities and contributes significantly to poverty reduction efforts. In Kenya, the sector provides jobs across value chains, generate revenue and contributes to food security for both rural and urban populations, making a significant economic contribution [1]. The livestock industry significantly impacts Kenya's food system, accounting for over 12% of the nation's Gross Domestic Product (GDP) and 22% of food system GDP [2], with the sector contributing 42% of agricultural GDP [3]. By 2050, Kenya's population is projected to reach an estimated 85 to 95 million, with close to half living in urban areas [4]. Due to population growth, urbanization, rising affluence, and shifting dietary preferences, demand for animal-source foods is expected to rise significantly [2]. Sustained increases in livestock production, effective management of feed and water resources, and evidence-based planning to support the livestock sector's sustainable development are all necessary to meet this rising demand. Therefore, evidence-based planning, strategic investment, and the development of a sustainable livestock sector depend on accurate forecasting frameworks that predict livestock populations, production, feed demand, and resource requirements. Statistical time-series models are often used to forecast animal numbers based on past data. These approaches anticipate future livestock numbers based on past population trends and have been used to aid in livestock planning and management. In [5], autoregressive and probabilistic time-series models were used to forecast pig populations, while [6] compared autoregressive integrated moving average (ARIMA) and artificial neural network (ANN) models for forecasting cattle populations in Türkiye. Similarly, [7] used exponential smoothing to forecast beef cattle population in Indonesia. While these approaches can effectively capture historical patterns and produce useful forecasts, they are mostly based on past trends and do not explicitly account for biological processes such as reproduction, mortality, maturation, culling, or offtake. As a result, their ability to assess the impact of biological characteristics and management interventions on future livestock population dynamics is restricted.

In contrast to statistical forecasting methods, biological simulation models represent the demographic dynamics that drive livestock populations. Animals are divided into cohorts based on their age, sex, or production stage, with transitions caused by biological processes such as reproduction, maturation, mortality, culling, and replacement. These models provide a biologically realistic picture of herd dynamics, allowing the evaluation of alternative management options. To simulate herd dynamics, [8] created an age- and sex-structured cattle population model that takes into account management techniques and animal migrations. Such models are useful decision-support tools because they link biological processes to herd structure and productivity. Existing biological models therefore provide valuable representations of herd dynamics, but a gap remains for replicable frameworks that jointly forecast population dynamics, production, feed requirements, and resource constraints across multiple livestock value chains. To address these shortcomings, this study developed a cohort-based quarterly forecasting framework for Kenya's nine major livestock value chains. The framework explicitly modeled biological and management processes, including reproduction, mortality, maturation, culling, and offtake, and links livestock population dynamics to species-specific production and feed demand. By integrating these components, the framework enabled consistent projections of livestock populations, production, and resource requirements, providing a decision-support tool for evidence-based livestock planning and policy development in Kenya.

2 Literature Review

2.1 Significance of the Livestock Sector in Kenya

Livestock production plays a key role in Kenya's economy and in household livelihoods, providing income, employment, wealth accumulation, and food. About 80% of Kenya's land area is classified as arid and semi-arid land (ASAL), where pastoralism is the primary source of income and carries about 75% of the national herd. Livestock production contributes an estimated 42% of agricultural GDP and around 12% of Kenya's total GDP [3].

National projections show that Kenya's population will continue to grow, reaching an estimated 85 to 95 million by 2050, with close to half living in urban areas [4]. This demographic and urban transition is expected to drive a substantial rise in demand for animal-source foods, creating new opportunities across the different livestock value chains [9].

2.2 Historical trends in livestock population and production

Livestock production historical trends also suggest a close association between changes in output and changes in livestock numbers, but not necessarily in proportional terms. At continental level, the African Union's Livestock Development Strategy for Africa states that increases in herd and flock size have contributed significantly to the growth in the supply of beef, milk, poultry and sheep and goat meat [10]. This finding suggests that population dynamics should be understood in the future assessment of livestock supply and that the growth of livestock populations may add to total production without being reflected in equivalent productivity increases.

Evidence from Kenya's smallholder dairy sector further illustrates the relationship between livestock population structure and production. Wanjala and Njehia [11] noted that poor performance of dairy herd was associated with inadequate replacement stock, limited access to improved breeding animals, inadequate feed quantity and quality, breed characteristics and short lactation periods. The results are dairy cattle specific, but they do suggest a substantial consideration for livestock population and production analysis in general.

Previous studies imply that forecasts based purely on aggregate historic growth rates may overlook changes that occur within livestock populations over time. A more disaggregated approach can therefore be a basis for studying how changes in livestock numbers translate into production and then into livestock feed requirements. This provides a basis for cohort-based forecasting of livestock populations, production, and feed demand across Kenya's livestock value chains.

2.3 Modelling Approaches for Livestock Population and Production Forecasting

Statistical modeling approaches have been used to explore factors that drive livestock productivity in Kenya. For instance, Ajak et al. [12] assessed dairy cattle productivity on smallholder farms in relation to factors such as feeding, breed, and disease, attributing poor performance largely to feed shortages alongside management- and price-related constraints. More recently, Were et al. [13] used multilevel mixed-effects linear regression to quantify the animal- and feeding-level factors associated with daily milk yield in smallholder herds in Kiambu County, identifying specific feeding practices and cow-level characteristics as significant predictors of output. Although such studies are primarily explanatory rather than predictive, they identify the production drivers that can be incorporated as explanatory variables in forecasting models.

2.4 Cohort-Based Herd Dynamics, Population and Production Projection Models

Empirical applications have highlighted the significance of explicitly considering the population structure of livestock. For example, Lesnoff et al. [14] used a demographic model to study the recovery of the cattle population after drought and showed that the recovery was particularly sensitive to reproductive performance. Likewise, Tuffa and Treydte [15] developed a stochastic age and sex-structured model for Boran cattle which included demographic processes, carrying capacity, market conditions and herder decisions. Their results suggested that decisions to remove animals from different demographic groups can drastically change population trajectories. Combined, these studies indicate that livestock population projections are more informative when they include the demographic processes by which populations are replenished, mature and removed, rather than just using aggregate growth rates.

2.5 Biological Transition Parameters: Mortality, Reproduction, Maturation and Off-take

Studies from across Sub-Saharan Africa have documented significant differences in mortality, fertility, maturation and off-take rates across livestock species and production environments. In a systematic review of livestock production parameters, Otte and Chilonda [16] summarized the relative high mortality of young cattle and small ruminants, low reproductive performance and generally low off-take in traditional production systems. Their synthesis also showed that population growth rates vary widely among species and production systems, emphasizing the necessity of biologically and geographically relevant transition parameters. Off-take is important especially as animals may leave the herd by sale or slaughter before the end of their biological life. These processes imply that the observed changes in livestock population are the net result of births, maturation, mortality and removals.

2.6 Feed Availability and Feed Demand

Availability of feed resources also limits the growth of the livestock population. Conventional livestock feed balance methods calculate how dry matter animals need by looking at their body weight or using livestock unit equivalents. Common intake rates like the 2.5 percent of body weight per day standard have been helpful for figuring out how much feed animals need when doing livestock assessments. Applications across Sub-Saharan Africa have used these approaches to identify substantial feed deficits. These developments suggest that livestock population and production projections can provide a useful basis for estimating future feed requirements.

2.7 Research Gap

The reviewed literature establishes the building blocks of the present study on solid methodological ground. Historical evidence links changes in livestock numbers to changes in output [10, 11]; statistical and regression-based studies identify feeding, and cow-level factors that drive productivity [12, 13]; demographic and cohort-based models show that population projections are far more informative when they represent how animals are born, mature, and are removed than when they rely on aggregate growth rates [14, 15]; syntheses of Sub-Saharan production parameters document the mortality, fertility, maturation, and off-take rates needed to drive such models [16]; and rangeland studies show that herd size must be matched to forage availability, which remote sensing now allows to be estimated across seasons and space [17].

What remains scarce is a single, reproducible, biologically grounded framework that unifies these strands for Kenya—one that simultaneously (i) spans all nine key livestock value chains rather than a single species, (ii) advances cohort transitions at a quarterly time step to capture seasonality, and (iii) closes the loop from population to species-specific production to disaggregated feed demand. The closest prior work is Tuffa and Treydte [15], whose stochastic age- and sex-structured model already couples cohort demography to carrying capacity and herder market decisions; the present study differs by trading that single-species depth for breadth, unifying nine value chains within one quarterly, parameter-transparent architecture, while leaving an equivalent carrying-capacity treatment as a deliberate scoping choice for future work (Section 6). By integrating these elements into one coherent architecture, the present study addresses a clear and consequential gap, offering not only defensible forecasts but a diagnostic instrument for identifying the feed deficiencies, low off-take rates, and reproductive bottlenecks that aggregate assessments obscure.

3 Methodology

3.1 Overview

Nine value chains (both ruminant and non-ruminant animals) were selected on the basis of their contribution to Kenya's agricultural GDP [1]. For each chain, forecasting covers three domains: **population dynamics**, **production volumes**, and **feed demand**. The baseline year is 2019; the projection horizon runs from 2020 to 2041.

A first-principles modelling approach was adopted. This choice reflects (i) the transparency and auditability it affords where each intermediate quantity is directly traceable to its inputs, and (ii) the heterogeneous, multi-source nature of the available data, which precludes the large homogeneous datasets required by machine-learning methods.

Parameters were compiled from peer-reviewed literature, government publications, and sector reports, then validated by livestock experts from the Ministry, associations, and farmers. Accepted values and their sources are published as a supplementary dataset [18].

The model operates on a quarterly time step. For each quarter, the three domains are computed in sequence.

3.2 Population Model Equations

Population is tracked across a defined cohort structure specific to each value chain. Within each quarter, the following processes are applied in order where relevant: mortality, culling, offtake, transition to the next age cohort, and births. The starting population for each cohort is updated by sequentially applying the corresponding rates, with inflows from younger cohorts and new births added at the end of the period.

3.2.1 Beef Cattle

$$P(t+1) = \left[P(t)\,(1-m_q)(1-k_q)(1-o_q)\right] + G(t) - E(t) + B(t) \tag{1}$$

$P(t)$	Starting pop.	m_q	Mortality rate (qtr.)
$P(t+1)$	Ending pop.	k_q	Culling rate
$G(t)$	Inflow (young cohort)	o_q	Offtake rate (steers)
$E(t)$	Outflow to next cohort	$B(t)$	New births (calves)

3.2.2 Goats

$$P(t+1) = \left[P(t)\,(1-m_q)(1-k_q)\right] + G(t) - E(t) + B(t) \tag{2}$$

Variable definitions follow the same convention as beef cattle; the offtake term o_q does not apply.

3.2.3 Camels

$$P(t+1) = P(t) - D(t) - C(t) - O(t) + B(t) \tag{3}$$

$D(t)$	Natural mortality	$C(t)$	Culling offtake
$O(t)$	Commercial offtake	$B(t)$	Births in period t

3.2.4 Poultry

Three sub-sectors are modelled separately. T_{bro} denotes annual broiler throughput and S_r the rearing survival rate.

$$P_{\text{ind}}(t+1) = P_{\text{ind}}(t) - E_{\text{ind}}(t) + R_{\text{ind}}(t) + A(t) \tag{4}$$

$$P_{\text{lay}}(t+1) = P_{\text{lay}}(t) - E_{\text{lay}}(t) + \text{DOC}(t) \times S_r \tag{5}$$

$$T_{\text{bro}}(t+1) = T_{\text{bro}}(t) + A(t) \tag{6}$$

The sector addition term $A(t)$ and its sigmoid growth modifier $g(t)$ are:

$$A(t) = P(t)\,(r_{\text{ent}} + r_{\text{exp}})\,g(t) \tag{7}$$

$$g(t) = g_{\min} + (g_{\max} - g_{\min})\,\exp\!\left(-\kappa^2\,\frac{(t-t_0)^2}{2}\right) \tag{8}$$

3.2.5 Apiculture

$$H(t+1) = H(t)\,(1-\delta(t))(1-\rho(t)) + S(t) \tag{9}$$

$\delta(t)$	Seasonal colony loss	$\rho(t)$	Queen replacement rate
$S(t)$	New colonies (swarming)		

3.2.6 Pigs

$$P(t+1) = P(t) - D(t) - C(t) - O(t) + B(t) + R(t) \tag{10}$$

where $R(t)$ represents replacement additions to the breeding herd. Variable definitions for $D(t)$, $C(t)$, $O(t)$, $B(t)$ follow the camel convention.

3.2.7 Rabbits

Rabbits follow the same population equation as pigs (including $R(t)$).

3.3 Production Model Equations

Production volumes are derived from the population state at the end of each quarter. Each value chain has a distinct production equation reflecting the commodity it yields, that is meat, milk, eggs, honey, or wax. For ruminants, production is computed separately for each cohort and then aggregated. For poultry, the three sub-sectors are modelled independently. Apiculture production is calculated at the hive level and then scaled by the number of hives.

3.3.1 Beef Cattle

$$\text{Beef (kg)} = \sum_c N_c \times \text{LW}_c \times \text{DP}_c \tag{11}$$

where $c \in \{\text{steers, cull cows, cull bulls}\}$, LW is live weight (kg) and DP is dressing percentage.

3.3.2 Goats

$$\text{Meat (kg)} = (N_{\text{removed}} + N_{\text{excess bucks}}) \times \text{LW} \times \text{DP} \tag{12}$$

$$\text{Milk (kg)} = N_{\text{does}} \times f_L \times Y_d \times D_L \tag{13}$$

N_{does}	Surviving does
f_L, Y_d, D_L	Lactating fraction; daily yield (kg); lactation days
$N_{\text{excess bucks}}$	Bucks above 4% herd cap, removed for slaughter

3.3.3 Camels

$$\text{Milk (kg yr}^{-1}) = N_{\text{adult females}} \times f_L \times Y_d \times 365 \tag{14}$$

$$\text{Meat (kg)} = \sum_c N_c \times \text{LW}_c \times \text{DP}_c \tag{15}$$

where $c \in$ {juv. offtake, adult offtake, cull adults}.

3.3.4 Poultry

$$\text{Eggs}_{\text{ind}}(t) = P(t) \times f_{\text{fem}} \times Y_e \tag{16}$$

$$\text{Eggs}_{\text{lay}}(t) = P(t) \times \eta_{\text{HD}} \times 365 \tag{17}$$

$$\text{Meat (kg)} = T_{\text{bro}} \times \text{LW} \times \text{DP} \tag{18}$$

where f_{fem} is female share, Y_e is eggs per hen per year, and η_{HD} is hen-day production rate.

3.3.5 Apiculture

$$\text{Honey}_{h,q} = H_h \cdot \frac{Y_h}{4} \cdot \bar{s} \cdot (1-\varphi) \cdot \Omega(t) \cdot \varepsilon_h \tag{19}$$

$$\text{Wax}_{h,q} = \text{Honey}_{h,q} \times \omega_h \tag{20}$$

$$\text{Nectar}_{h,q} = H_h \cdot \frac{N_h}{4} \cdot \bar{s} \cdot \phi_q \cdot \nu_q \cdot \Omega(t) \tag{21}$$

Y_h, $\bar{s}$	Annual yield/hive (kg); effective colony strength
φ	Forage competition penalty (0–0.20)
ε_h	Hive-type efficiency factor
$\Omega(t)$	Technology/training improvement (sigmoid)
ω_h	Wax-to-honey ratio
ϕ_q, ν_q	Quarterly forage and nectar seasonality factors

3.3.6 Pigs and Rabbits

$$\text{Meat} = N_{\text{slaughtered}} \times \text{LW} \times \text{DP} \tag{22}$$

3.4 Feed Model Equations

Feed demand is estimated using dry matter intake (DMI) for all value chains. A DMI-based approach was adopted due to limited availability of detailed ration composition data across most value chains; it provides a consistent and tractable basis for estimating feed requirements from population and live weight alone. The specific formulations vary by value chain, as detailed below.

3.4.1 Beef Cattle, Pigs, and Rabbits

All three value chains use the same daily dry matter intake (DMI) formulation:

$$\mathrm{DMI} = P(t) \times \mathrm{LW} \times r_{\mathrm{DMI}} \times 365 \tag{23}$$

where r_{DMI} is daily dry matter intake as a proportion of body weight.

3.4.2 Goats

Feed is summed across cohorts (kids, weaners, growers, does, bucks):

$$\mathrm{Feed}(t) = \sum_{\mathrm{cohort}} N_c \times \mathrm{BW}_c \times r_{\mathrm{DMI},c} \times D \tag{24}$$

where D is the number of days in the period.

3.4.3 Camels

Feed is accumulated quarterly, with intake weighted by average live weight over the quarter:

$$\text{Annual feed} = \sum_{q=1}^{4} \mathrm{Feed}_q \tag{25}$$

$$\mathrm{Feed}_q = \sum_{\mathrm{cohort}} N_c \times \left(\frac{\mathrm{BW}_{0,c} + \mathrm{ADG}_c \times 90}{2} \right) \times r_{\mathrm{DMI},c} \times 90 \tag{26}$$

where ADG is average daily gain ($\mathrm{kg\,d^{-1}}$).

3.4.4 Poultry

Each sub-sector has a distinct feed equation reflecting its production cycle structure:

$$\begin{aligned} F_{\mathrm{ind}}(t) &= \big[R(t)(f_{\mathrm{chk}}\, d_{\mathrm{chk}} + f_{\mathrm{gr}}\, d_{\mathrm{gr}}) \\ &\quad + P(t)\, f_{\mathrm{ad}}\, 365\big](1-\sigma)^t && (27) \\ F_{\mathrm{lay}}(t) &= \mathrm{DOC}(t)(f_{\mathrm{chk}}\, d_{\mathrm{chk}} + f_{\mathrm{pul}}\, d_{\mathrm{pul}}) \\ &\quad + P(t)\, f_{\mathrm{hen}}\, 365 && (28) \\ F_{\mathrm{bro}} &= N_{\mathrm{birds}} \sum_{\mathrm{phase}} f_{\mathrm{phase}} \times d_{\mathrm{phase}} \times C_{\mathrm{yr}} && (29) \end{aligned}$$

where $f_{(\cdot)}$ is phase-specific intake ($\mathrm{kg\,d^{-1}}$), $d_{(\cdot)}$ is phase duration (days), σ is the scavenging offset, and C_{yr} is production cycles per year.

4 Results and Discussion

Population and production are reported over 2019–2041. The 2019 values are historical observations; 2020–2041 are model projections. The 2019 baseline is therefore shown shaded in the figures, with the 2019–2020 segment drawn dotted, and where the transition from historical to projected data produces a step, growth rates are quoted from 2020. Feed demand carries no 2019 historical value and is reported over 2020–2041 throughout.

4.1 Population

4.1.1 Ruminants

Total ruminant numbers rise from 88.0 million head in 2019 to 272.1 million in 2041, a 3.1-fold increase equivalent to 5.3 % per year (Table 1, Figure 1). Growth is not evenly distributed across value chains. Beef cattle expand fastest at 6.6 % per year, quadrupling to 67.8 million head and overtaking sheep as the second-largest ruminant population in 2036. Goats remain the largest single population throughout, growing 5.9 % per year to 121.3 million. Camels and dairy cattle follow intermediate trajectories of 5.2 % and 4.9 % per year respectively, each roughly tripling. Sheep are the slowest-growing ruminant chain at 3.3 % per year, doubling over the period; their share of the ruminant herd falls from 31.2 % in 2019 to 20.5 % in 2041.

Table 1 Ruminant population, 2019–2041 (thousand head)

Series (000 head)	**2019**	**2025**	**2030**	**2035**	**2041**	**CAGR 2019–41 (%)**
Goats	34 658	50 854	66 733	87 571	121 333	5.9
Beef cattle	16 640	23 168	32 365	45 282	67 767	6.6
Sheep	27 441	33 099	39 221	46 098	55 665	3.3
Camels	4 722	6 661	8 535	10 794	14 265	5.2
Dairy cattle	4 537	5 929	7 593	9 726	13 091	4.9
Total ruminants	87 998	119 710	154 448	199 470	272 119	5.3

4.1.2 Non-ruminants

Non-ruminant numbers grow far more slowly, from 106.3 million head in 2019 to 159.1 million in 2041, a 1.5-fold increase at 1.8 % per year (Table 2, Figure 2). The aggregate conceals a divergence between the commercial and non-commercial segments of the poultry chain. Layers and broilers expand by only about 10 % over 22 years (0.4 % per year) and plateauing after the early 2030s, whereas indigenous poultry grow steadily at 3.0 % per year to 89.3 million birds. The composition of the poultry flock shifts accordingly: indigenous birds rise from 44.1 % to 58.1 % of total poultry, while the combined layer and broiler share of all non-ruminants falls from 55.2 % to 40.5 %. The smallest chains grow fastest: rabbits 4.4-fold at 7.0 % per year and pigs 3.5-fold at 5.9 % per year. Both start below one million head, so they remain marginal in absolute terms. Managed bee colonies increase 2.7-fold to 3.5 million.

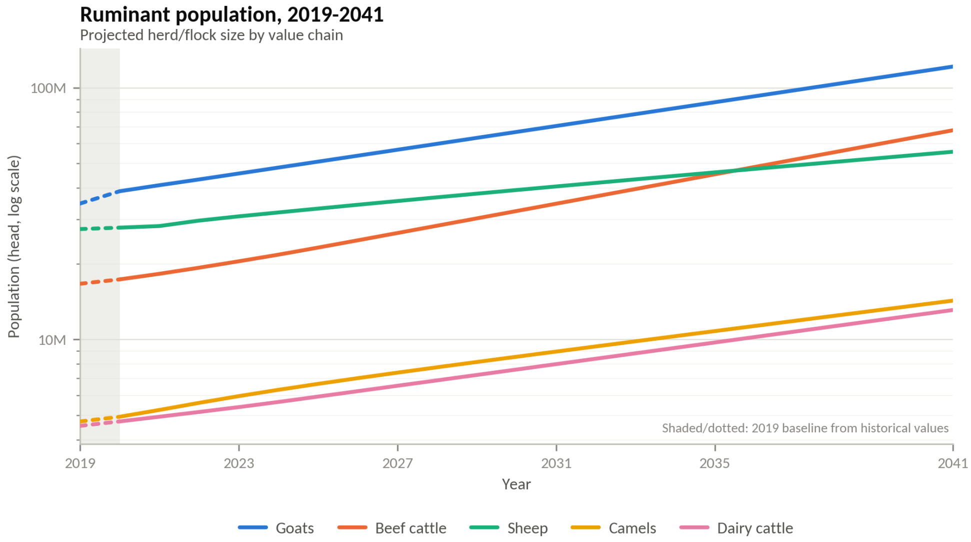

Fig. 1 Projected ruminant population by value chain, 2019–2041. Logarithmic vertical axis; the shaded band and dotted segment mark the 2019 baseline, which is a historical value.

Table 2 Non-ruminant population, 2019–2041 (thousands). Bee colonies are reported separately and excluded from the total.

Series (000)	**2019**	**2025**	**2030**	**2035**	**2041**	**CAGR 2019–41 (%)**
Indigenous poultry	46 293	54 774	65 960	75 970	89 288	3.0
Broilers	45 838	47 477	49 975	50 175	50 386	0.4
Layers	12 844	13 114	13 946	14 030	14 089	0.4
Rabbits	737	1 149	1 585	2 191	3 235	7.0
Pigs	596	817	1 097	1 471	2 088	5.9
Total (excl. colonies)	106 309	117 331	132 563	143 837	159 086	1.8
Bee colonies	1 289	1 838	2 384	2 926	3 489	4.6

4.2 Production

4.2.1 Ruminants

Ruminant output expands across every product line (Table 3, Figure 3). Ten of the reported series carry a step between the 2019 historical value and the 2020 projection; for those series growth is described here from 2020, and the 2019–2041 CAGRs in Table 3 should be read with that caveat.

Total milk from cattle, camels, and goats rises from 5.36 billion kg in 2020 to 15.41 billion kg in 2041, 5.2 % per year. Cattle milk remains the largest single product at 9.26 billion kg in 2041; camel milk is the second largest at 5.61 billion kg, and since the two grow at almost identical rates from 2020 (5.0 % and 5.3 % per year) their ratio is close to stable, narrowing only from 1.75:1 to 1.65:1. Goat milk is the fastest-growing dairy product at 6.6 % per year but remains a lot less at 531 million kg.

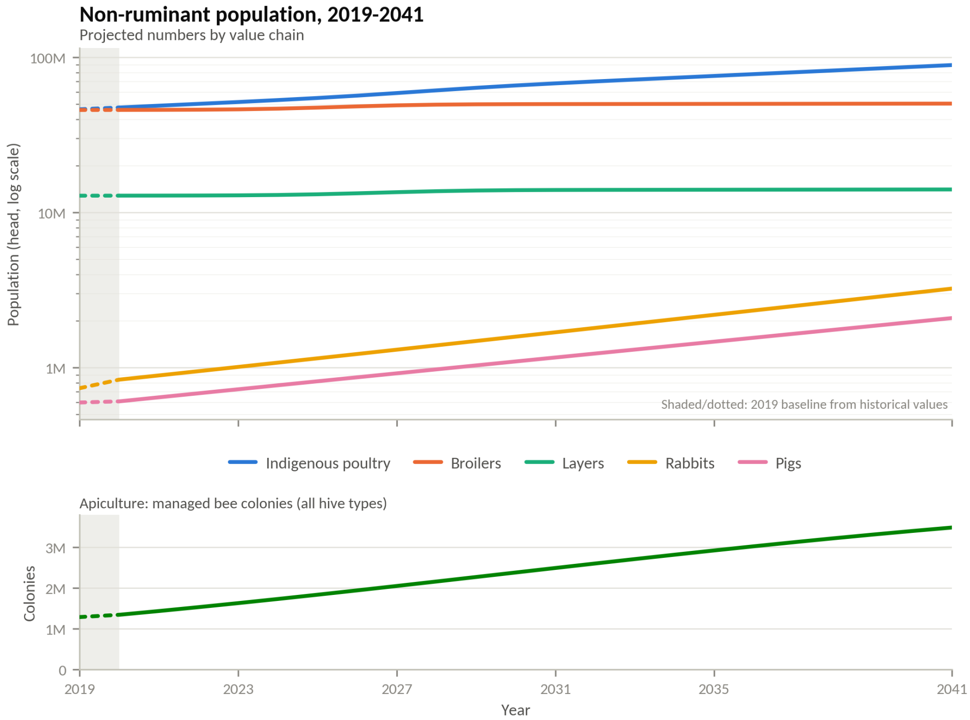

Fig. 2 Projected non-ruminant population by value chain, 2019–2041. Upper panel: livestock numbers (logarithmic axis). Lower panel: managed bee colonies (linear axis). Shaded bands and dotted segments mark the 2019 baseline, which is a historical value.

Meat output is led by chevon, which overtakes beef in 2020 and reaches 1.71 billion kg by 2041 against 1.37 billion kg of beef; both grow at 5.3–5.5 % per year from 2020. Camel meat is the fastest-growing ruminant meat at 6.8 % per year, though at 235 million kg in 2041 it remains well behind the two leading products. Mutton grows slowest of all ruminant products at 1.2 % per year from 2020, reaching 165 million kg, so the sheep chain contributes a declining share of ruminant meat. Wool, the only product in the third group, reaches 3.9 million kg at 2.8 % per year.

Table 3 Ruminant production, 2019–2041 (million kg)

Series (million kg)	**2019**	**2025**	**2030**	**2035**	**2041**	**CAGR 2019–41 (%)**
Cattle milk	3 087	4 205	5 375	6 883	9 263	5.1
Camel milk	985	2 672	3 369	4 248	5 611	8.2
Goat milk	156	222	292	383	531	5.7
Chevon	53.9	718	942	1 236	1 712	17.0
Beef	322	479	656	919	1 375	6.8
Camel meat	57.6	104	139	178	235	6.6
Mutton	49.4	100	117	137	165	5.6
Wool	0.84	2.40	2.80	3.26	3.92	7.3

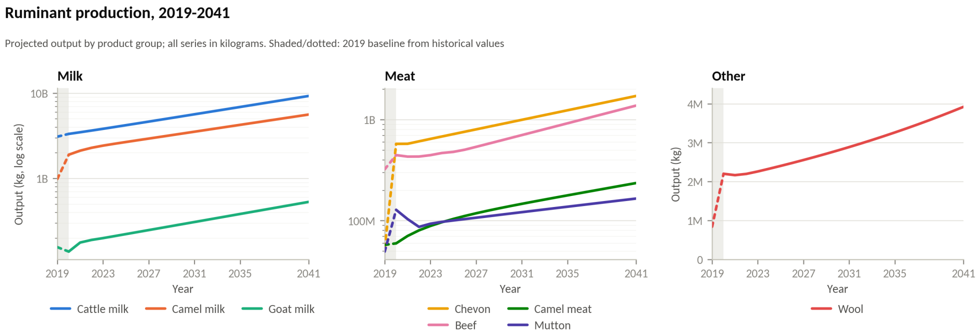

Fig. 3 Projected ruminant production, 2019–2041, by product group: milk, meat, and wool. Milk and meat panels use logarithmic vertical axes; the wool panel is linear. Shaded bands and dotted segments mark the 2019 baseline, which is a historical value.

4.2.2 Non-ruminants

Non-ruminant production growth is more subdued and closely tracks the population trajectories (Table 4, Figure 4). Poultry meat, the largest non-ruminant product, grows only 1.3 % per year to 109.7 million kg, consistent with the plateau in commercial bird numbers. Pork grows at 4.0 % per year to 34.1 million kg and rabbit meat at 4.6 % per year (from 2020) to 4.2 million kg. The apiculture products grow fastest: honey at 5.1 % per year to 41.6 million kg and beeswax at 5.4 % per year (from 2020) to 6.5 million kg. Honey output exceeds pork output in absolute mass from 2021 onward and is 1.2 times pork by 2041, making apiculture the second-largest non-ruminant product group by mass. Egg production, reported in trays and therefore shown in its own panel on a linear axis, rises from 180.4 million trays in 2020 to 228.7 million in 2041 (1.1 % per year); the 501.3 million trays recorded for 2019 is inconsistent with the remainder of the series.

Table 4 Non-ruminant production, 2019–2041 (million kg; eggs in million trays)

Series (million)	**2019**	**2025**	**2030**	**2035**	**2041**	**CAGR 2019–41 (%)**
Poultry meat	82.4	88.8	97.6	103	110	1.3
Pork	14.4	18.2	22.2	27.0	34.1	4.0
Rabbit meat	1.76	2.03	2.55	3.19	4.18	4.0
Honey	13.9	22.4	29.2	35.4	41.6	5.1
Beeswax	1.37	3.39	4.48	5.50	6.54	7.4
Eggs (million trays)	501	189	208	217	229	-3.5

4.3 Feed demand

4.3.1 Ruminants

Ruminant chains account for the great majority of projected feed demand (Table 5, Figure 5). Total requirement rises from 105.8 million tonnes DM in 2020 to 353.8 million tonnes in 2041, 5.9 % per year. Beef cattle alone account for 58.8 million tonnes in 2020 and 218.3 million in 2041, more than the other four ruminant chains combined

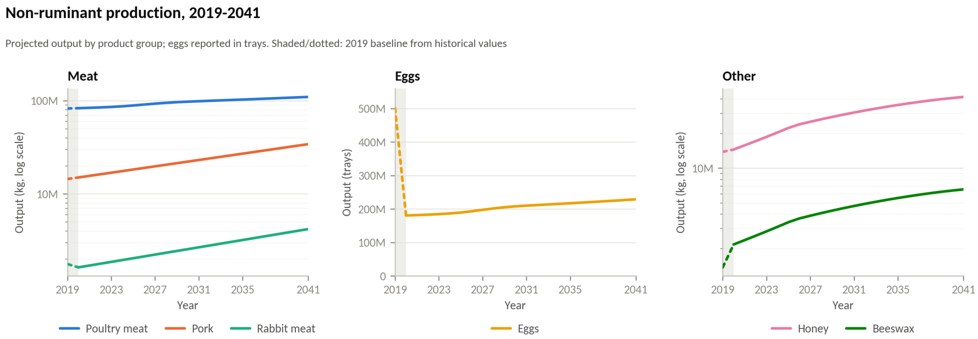

Fig. 4 Projected non-ruminant production, 2019–2041, by product group: meat, eggs, and apiculture products. Meat and other panels use logarithmic vertical axes; the egg panel is linear because eggs are reported in trays. Shaded bands and dotted segments mark the 2019 baseline, which is a historical value.

in every year of the period. Their share rises from 55.5 % to 61.7 % as they grow at 6.4 % per year, the fastest of the five chains. Goats follow at 6.1 % per year to 36.2 million tonnes and camels at 5.6 % to 42.2 million tonnes, the latter overtaking dairy cattle in 2023; dairy cattle reach 41.2 million tonnes at 4.9 % per year. Sheep demand grows slowest, from 8.0 to 15.9 million tonnes at 3.3 % per year.

Table 5 Ruminant feed demand, 2020–2041 (thousand tonnes DM)

Series (000 t DM)	**2020**	**2025**	**2030**	**2035**	**2041**	**CAGR 2020–41 (%)**
Beef cattle	58 791	74 615	104 258	145 888	218 333	6.4
Camels	13 474	19 399	25 167	31 904	42 184	5.6
Dairy cattle	15 064	18 646	23 877	30 584	41 165	4.9
Goats	10 528	15 187	19 929	26 151	36 233	6.1
Sheep	7 989	9 332	11 135	13 133	15 889	3.3
Total ruminants	105 846	137 177	184 366	247 660	353 805	5.9

4.3.2 Non-ruminants

Non-ruminant feed demand is less, rising from 3.29 million tonnes in 2020 to 6.46 million tonnes in 2041 at 3.3 % per year (Table 6, Figure 6). That is 1.8 % of the ruminant total in 2041. Poultry accounts for the largest share throughout but grows slowest at 1.9 % per year to 2.83 million tonnes, so its share falls from 57.7 % to 43.8 %. Apiculture nectar demand grows at 5.0 % per year to 2.49 million tonnes, reaching 88 % of poultry feed demand by 2041. Pig feed demand reaches 1.09 million tonnes (4.1 % per year) and rabbit demand 52.7 thousand tonnes (4.5 % per year).

5 Managerial, Policy, and Societal Implications

Beyond its forecasting function, the cohort-based framework developed in this study operates as a diagnostic instrument that translates biological herd dynamics into

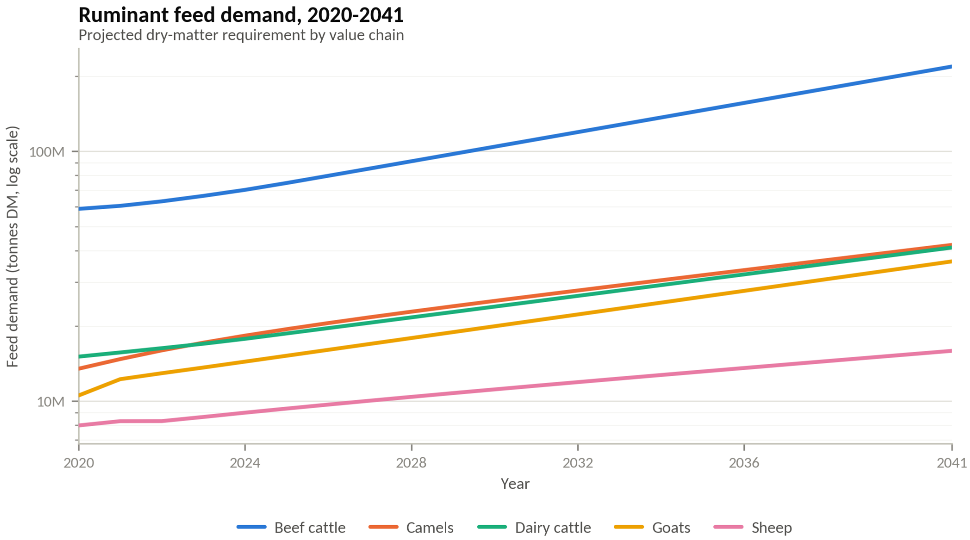

Fig. 5 Projected ruminant dry-matter feed demand by value chain, 2020–2041. Logarithmic vertical axis.

Table 6 Non-ruminant feed demand, 2020–2041 (thousand tonnes)

Series (000 t)	**2020**	**2025**	**2030**	**2035**	**2041**	**CAGR 2020–41 (%)**
Poultry	1 900	2 067	2 350	2 556	2 828	1.9
Apiculture (nectar)	900	1 259	1 706	2 093	2 489	5.0
Pigs	473	579	707	862	1 092	4.1
Rabbits	20.9	25.8	32.2	40.3	52.7	4.5
Total non-ruminants	3 293	3 931	4 795	5 552	6 462	3.3

decision-driven intelligence. By resolving each value chain into age and sex-structured cohorts advanced on a quarterly basis, its outputs speak simultaneously to herd and enterprise managers, to national and county policymakers, and to the broader societal objectives of food security, livelihood resilience, and environmental sustainability.

5.1 Managerial Implications

At the enterprise level, the framework identifies precisely where within a production system growth is being lost by isolating the cohorts and transition processes that constrain performance—e.g., elevated calf, lamb, and kid mortality, depressed reproductive rates, and sub-optimal off-take structures. This granularity matters because, as seen from comparable East African rangeland systems indicates, the sale and retention rates of breeding and mature females exert the greatest influence on long-run population trajectories [15, 19].

Managers may therefore use model outputs to assess herd structure, inform culling and destocking decisions, and breeding interventions. The quarterly time step also

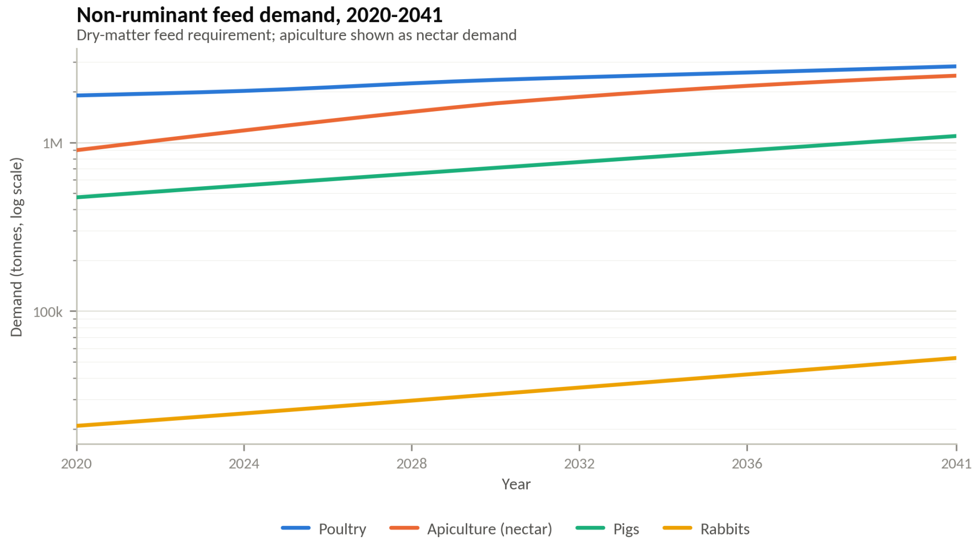

Fig. 6 Projected non-ruminant feed demand by value chain, 2020–2041. Apiculture is shown as nectar demand. Logarithmic vertical axis.

facilitates monitoring seasonal changes in herd dynamics and feed requirements, potentially supporting early intervention where projected feed deficits or changes in herd condition indicate emerging risks [17].

5.2 Policy Implications

For national and county governments the framework directly translates into the analytical needs of national and county planning providing the value chain specific projections necessary to prioritize investment and set performance targets. The ability to run coherent forward scenarios across nine value chains provides a basis for evidence-based investment sequencing and could be institutionalized within national statistical and livestock agencies, transforming sector planning from static census snapshots to continuous data-driven foresight [20].

5.3 Societal Implications

At the societal scale, the framework addresses the demand-driven transformation of animal-source-food consumption that will accompany Kenya's population growth and urbanization. Reliable projections of milk, meat, egg, and honey supply against rising demand are central to food and nutrition security planning, allowing anticipatory action on emerging supply gaps rather than reactive crisis response. Since livestock underpins the livelihoods of several million pastoralists and agro-pastoralists and constitutes the dominant source of income across the arid and semi-arid lands, the model's capacity to anticipate feed shortfalls and productivity shocks has direct implications for poverty reduction, resilience, and the protection of vulnerable rural households [1].

6 Limitations and Opportunities for Future Research

The model does not currently incorporate environmental carrying capacity, so projected populations are not checked against the forage, land, or water base that would ultimately constrain them. Existing carrying-capacity methods diverge substantially in both approach and result: zone-based partitioning of agro-ecological data, remote-sensing-derived stocking rates [17], and field-sampled biomass estimates can each be applied to the same rangeland and yield different sustainable-offtake thresholds, with sensitivity analysis showing swings of up to 20 percentage points in the area classified as overutilised depending on the modelling assumptions used [21]. Population models that do embed carrying capacity, such as the stochastic Boran cattle model of Tuffa and Treydte [15], treat its estimation as a substantial undertaking in its own right rather than a parameter that can be added on; integrating it here, ideally partitioned by agro-ecological zone and value chain, is better scoped as an independent follow-on study.

Data availability also constrained several value chains. Camel, goat, and sheep milk are frequently reported in Kenyan statistics as a combined "other milk" category rather than disaggregated by species, so chain-specific dairy parameters could not always be pinned to a domestic figure. For parameters with little or no Kenyan data, values were instead drawn from studies conducted in regions with comparable climate and breeds, a substitution consistent with FAO's own finding that livestock production parameters vary widely across production systems and geographies in Sub-Saharan Africa [16], and one that introduces uncertainty proportional to how closely the donor region matches Kenyan conditions. Camel and apiculture parameters were the hardest to source domestically and lean most heavily on this substitution, so their 2041 projections should be treated as the least certain in the model; a formal sensitivity analysis quantifying how much the borrowed parameters move the headline growth rates has not yet been carried out and is a natural next step ahead of the carrying-capacity work above.

A further limitation not covered by the data issues above is scope: because the model is first-principles and driven by biological transition rates, it does not represent exogenous shocks that can move herd trajectories independently of demography, chief among them disease outbreaks, market price swings, and policy changes. Structured population models elsewhere in the literature that couple demographic transitions to herder market decisions, such as Tuffa and Treydte [15], show that offtake responses to these external drivers can materially alter population trajectories; their omission here means the projections should be read as a demographic baseline rather than a forecast robust to shocks.

Addressing these gaps points to five priorities for future work: (i) a dedicated carrying-capacity study using zone- and value-chain-specific partitioning; (ii) targeted primary data collection for the species and parameters most affected by aggregation or missing coverage; (iii) integration of economic analysis, including price and cost feedbacks; (iv) incorporation of exogenous drivers such as disease, markets, and policy, potentially via a hybrid first-principles/machine-learning extension; and (v) disaggregating each value chain into independent, and where relevant breed-specific, sub-models for finer-grained decision support.

7 Conclusion

This study set out to address a persistent gap in Kenya's livestock sector planning: the absence of a reproducible, data-driven framework linking herd demographics, production, and feed requirements across value chains. The resulting model resolves nine value chains, dairy cattle, beef cattle, camels, goats, sheep, poultry, pigs, rabbits, and apiculture, into age- and sex-structured cohorts advanced on a quarterly step from a 2019 baseline through 2041, translating mortality, culling, offtake, maturation, and reproduction into population, production, and feed demand trajectories for each chain.

The projections point to sustained, uneven growth across the sector. Ruminant numbers roughly triple over the horizon, led by beef cattle and goats, while non-ruminant growth is dominated by the divergence between a plateauing commercial poultry segment and a steadily expanding indigenous flock; production and feed demand follow correspondingly divergent paths by value chain. Beyond these headline trajectories, the model's cohort-level resolution surfaces the specific structural constraints, such as mortality concentrated in young cohorts, low off-take, and uneven reproductive performance, that drive them, offering a diagnostic view that aggregate, top-down projections cannot provide.

This combination of forecast and diagnosis is what gives the framework its practical value. Enterprise managers can use it to identify where herd performance is being lost and target culling, breeding, or feeding interventions accordingly; national and county planners can use it to move from static census snapshots to continuous, value-chain-specific investment and policy planning; and at the sector level, it offers a basis for anticipating feed and supply gaps before they become food security risks, particularly for the pastoralist and agro-pastoralist communities most exposed to them.

These contributions should be read alongside the limitations discussed above, chiefly the exclusion of environmental carrying capacity, gaps in Kenya-specific data for several value chains, and the omission of exogenous shocks such as disease and market fluctuations. Closing these gaps, through dedicated carrying-capacity work, targeted data collection, and extensions that incorporate economic and exogenous drivers, is the natural next step for turning this framework into a fuller decision-support tool for Kenya's livestock sector.

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